Manuscript Figures for the Glioma Arm

Your `generate_figures1.py` specification, run against the real evidence. Four figures produced, one not producible, two panels changed for substantive reasons.

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Dear Colleague,

Thank you for the figure script — the specification came through clearly and that was the useful part. I reimplemented it against the actual evidence files rather than patching it, because a few of its data paths pointed at columns that do not exist in our outputs, and the fallbacks would have produced figures that looked right and were not. Everything below is generated by `evidence/m12_manuscript_figures.py` under 15 integrity gates, all passing; your original is preserved at `evidence/generate_figures1_AS_RECEIVED.py`.

1 What you have

Figure	Status	Content
1	produced	cohort sizes; marker prevalence; **real** scanner distribution; median OS
3	produced	IDH (TCGA, n=919); MGMT (UPENN, n=247); radiomic risk tertiles (UPENN, n=574)
4	not producible	habitat features have never been computed
5	produced, restructured	imaging vs molecular, like-for-like
Calibration	produced	out-of-fold, $\log_{10}$ on both axes

All three Kaplan–Meier panels are strongly positive and reproduce the earlier reports exactly: IDH 89.7 vs 14.0 months ($p = 5.7 \times 10^{-71}$), MGMT 595 vs 371 days ($p = 5.8 \times 10^{-8}$), and radiomic risk tertiles 490 vs 268 days ($p = 3.3 \times 10^{-8}$).

2 Two changes that affect what the manuscript can claim

These are not cosmetic, so I want to flag them explicitly rather than let them pass in a figure.

1. There is no external validation, so that bar is gone. The specification had "Imaging (external), C-index 0.581, $p = 0.012$". No such result exists anywhere in our work. R02 states the position directly: the 337-patient cohort intended for external validation *has no survival data at all*. It carries sex, age, MGMT, field strength and scanner model only. Figure 5 therefore reports the internally validated C-index (0.602, 95% CI 0.576–0.629) and nothing else. If the manuscript currently claims external validation, that claim needs to come out or a new cohort needs to be found.

2. Figure 5 has been restructured, because the comparison as specified is the one we withdrew. Placing "Molecular (all grades) 0.719" beside "Imaging 0.602" is exactly the grade-mix comparison R04's audit retracted: TCGA spans grades 2–4 while UPENN is glioblastoma only, and IDH status largely *is* the grade distinction. Restricted to glioblastoma the molecular panel falls to 0.541–0.561, i.e. *below* imaging. Panel A now shows that like-for-like comparison; panel B shows the naive version explicitly marked as invalid, since it is worth showing why rather than silently omitting it.

3 Three smaller corrections

3. Scanner distribution is real, but it is a different cohort and a different shape. The specification hard-coded $[443, 350, 267, 0]$ — roughly an even three-way split summing to 1,060, which matches no cohort we hold. The eight-hospital set is OpenNeuro ds007045 with **337** participants, and it is Siemens-dominated: **Siemens 228 (68%), Philips 89 (26%), GE 20 (6%)** across 21 distinct scanner models, at 3 T (227), 1.5 T (109) and 1 T (1). If the manuscript discusses multi-centre heterogeneity, the real distribution is considerably less balanced than the placeholder implied.

4. Median overall survival must be Kaplan–Meier, not a median of observed times. TCGA is 50% censored. A raw median reads 11.8 months; the Kaplan–Meier median is **20.7**. Using the raw figure would have made pan-glioma appear to survive *worse* than glioblastoma-only UPENN (12.8 months), which is backwards. UPENN is uncensored — all patients deceased — so its raw and Kaplan–Meier medians coincide, which is why the two are not interchangeable across cohorts. Panel D uses Kaplan–Meier throughout and annotates the difference.

5. The calibration plot is $\log_{10}$ on both axes. The specification compared our $\log_{10}$ predictions against `np.log` of observed survival, which mis-scales the diagonal by $\ln(10) \approx 2.30$.

4 Figure 4 cannot be produced

Habitat features have never been computed. Producing them is not an analysis of data in hand: it requires the image preprocessing pipeline — co-registration of four modalities, skull-stripping, resampling across 30+ distinct voxel grids — over roughly 967 patients. That is the one arm of this programme still unstarted, and it needs its own scope and cost estimate rather than a figure call. I have included a placeholder panel stating this, so the gap is visible in the figure set rather than silently absent.

5 Two notes on the script itself, offered as useful rather than critical

Both would have produced plausible-looking figures from wrong inputs, so they seem worth recording.

IDH would have been random. `load_tcga_data` reads `IDH_STATUS` from `data_clinical_patient.txt`, where the column does not exist — it lives in `data_clinical_sample.txt`. With no `patient_id` in our `tcga_survival.csv` either, control reaches the `np.random.choice` fallback. Run as written, Figure 3A would have plotted randomly assigned IDH labels and printed $p = 0.325$ with medians 12.2 vs 11.6 months — asserting that IDH is not prognostic in glioma. The join here is by `PATIENT_ID` from the sample file, there is no fallback, and a gate requires the result to reproduce 89.7 vs 14.0 months.

The radiomic "prediction" would have been the outcome. The script takes `oof.iloc[:, 1]`; in `oof_survival.csv` the columns are `survival_days, log10_days, predicted_log10_days`, so index 1 is the observed outcome. Figure 3C would have stratified survival by survival — $p \approx 10^{-104}$, spectacular and circular. Selecting by name gives the genuine result, $p = 3.3 \times 10^{-8}$. (Relatedly, the tertile labels were inverted: higher predicted log-survival is *lower* risk.)

I mention these only because the same two patterns — a silent fallback, and a column chosen by position — are easy to carry into any reimplementation, so both are now gated.

6 Reproducing this

- `evidence/m12_manuscript_figures.py` — one command, regenerates all five PDFs

- `evidence/gates.json` — 15 gates with expected/observed/PASS for each
- `evidence/results.json` — every number plotted, unrounded
- `evidence/config.yaml` — the plan, hashed before any figure was drawn (`084e0f49...47ac4ee`)
- `figures/` — PDF and PNG at 200 dpi

Happy to add panels, change the framing of Figure 5, or produce a supplementary version showing the all-grades comparison in full if you would rather argue that case explicitly. The one thing I would not do is restore the external-validation bar without a cohort behind it.

Kind regards,

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